

Dr. Bernat Font

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b-fg.github.io

Research interests

computational fluid dynamics; machine learning; turbulence modelling; numerical methods; high-performance computing.

Experience

Assistant Professor, Ship Hydromechanics section, Technische Universiteit Delft, Netherlands. 2024–

Focus: Data-Informed CFD, i.e. data exploitation to improve current CFD methods:

- Acceleration of scale-resolving simulations using generative methods
- Data-driven wall models for non-equilibrium turbulent boundary layers
- Active flow control using reinforcement learning
- Discovery of governing equations from raw data

Postdoctoral Researcher, Large-scale Computational Fluid Dynamics group, Barcelona Super-computing Center, Spain. 2021–2023

Topics:

- NextSim EU project – Next generation of industrial aerodynamic simulation code
- Machine learning for CFD: wall models and active flow control
- Turbulence modelling for high-order scale-resolving simulations

Researcher, Mathematical Institute, Oxford University, UK. 2020–2021

Topic: Transport of porous particles in fluid flow.

Visiting PhD Researcher, Institute of High-Performance Computing, A*STAR, Singapore. 2017–2020

Education

Ph.D. Computational Fluid Dynamics, University of Southampton, UK. 2015–2021

Thesis: Modelling of Flow Past Long Cylindrical Structures. ([eprint](#))

Supervisors: Prof. G.D. Weymouth, Prof. O.R. Tutty, Dr. V.-T. Nguyen.

Visiting Researcher: Research attachment funded by the ARAP mobility scheme, Institute of High-Performance Computing, A*STAR, Singapore.

M.Sc. Computational Fluid Dynamics, Cranfield University, UK. 2014–2015

Thesis: High-order Shock-capturing Schemes for Micro Shock Tubes. ([eprint](#))

Supervisor: Dr. L. Könözy.

Double Degree with Enginyeria Superior in Aeronautical Engineering.

Enginyeria Superior Aeronautical Engineering, Universitat Politècnica de Catalunya, Spain. 2012–2015

Mentor: Prof. C.-D. Pérez-Segarra.

Equivalent to Master of Engineering.

Enginyeria Tècnica Aeronautical Engineering, Universitat Politècnica de Catalunya, Spain. 2009–2012

Equivalent to Bachelor of Engineering.

Peer-reviewed journal articles

17. Montalà, R., Font, B., Suárez, P., Rabault, J., Lehmkuhl, O., Vinuesa, R., and Rodriguez, I. (2026, submitted). Towards practical deep-reinforcement-learning flow control in wings: Sensor placement effects on turbulent separation suppression. ([eprint](#))
16. Chillón, E., Lidtke, A.K., Doan, N.A.K., and Font, B. (2026, submitted). Acceleration of an algebraic multigrid pressure solver using graph neural networks. ([doi](#), [eprint](#))
15. Montalà, R., Font, B., Suárez, P., Rabault, J., Lehmkuhl, O., Vinuesa, R., and Rodriguez, I. (2026). Flow separation control of an infinite wing section via multi-agent reinforcement learning. *Journal of Fluid Mechanics*, 1040:A3. ([doi](#), [eprint](#))
14. Rubio, G., Ntoukas, G., Chávez-Módena, M., Mariño, O., Font, B., Lehmkuhl, O., Valero, E., and Ferrer, E. (2025, submitted). Can explicit subgrid models enhance implicit LES simulations? a GPU-oriented high-order-solver perspective. ([doi](#), [eprint](#))
13. Sababha, H., Font, B., and Daqaq, M. (2026). Deep reinforcement learning in action: Real-time control of vortex-induced vibrations. *Physics of Fluids*, 38(1):014102. ([doi](#), [eprint](#))
12. Xiao, M., Wang, Y., Rodach, F., Font, B., Kurz, M., et al. (2025, submitted). SmartFlow: A CFD-solver-agnostic deep reinforcement learning framework for computational fluid dynamics on HPC platforms. ([doi](#), [eprint](#))
11. Weymouth, G.D. and Font, B. (2025). Waterlily.jl: A differentiable and backend-agnostic julia solver for incompressible viscous flow around dynamic bodies. *Computer Physics Communications*, 315:109748. ([doi](#), [eprint](#))
10. Gao, M., Kutz, J.N., and Font, B. (2025, submitted). Mesh-free sparse identification of nonlinear dynamics. ([doi](#), [eprint](#))
9. Suárez, P., Alcántara Ávila, F., Rabault, J., Miró, A., Font, B., Lehmkuhl, O., and Vinuesa, R. (2025). Active flow control for drag reduction through multi-agent reinforcement learning on a turbulent cylinder at $Re_D = 3900$. *Flow, Turbulence and Combustion*, 115(1):3–27. ([doi](#), [eprint](#))
8. Garcia, X., Miró, A., Suárez, P., Alcántara Ávila, F., Rabault, J., Font, B., Lehmkuhl, O., and Vinuesa, R. (2025). Deep-reinforcement-learning-based separation control in a two-dimensional airfoil. *International Journal of Heat and Fluid Flow*, 116:109913. ([doi](#), [eprint](#))
7. Cutz, L., Misar, S., Font, B., Al-Naji, M., and de Jong, W. (2025). Hydrothermal liquefaction of spanish crude olive pomace for biofuel and biochar production. *Journal of Analytical and Applied Pyrolysis*, 188:107050. ([doi](#), [eprint](#))
6. Font, B., Alcántara-Ávila, F., Rabault, J., Vinuesa, R., and Lehmkuhl, O. (2025). Deep reinforcement learning for active flow control in a turbulent separation bubble. *Nature Communications*, 16(1):1422. ([doi](#), [eprint](#))
5. Suárez, P., Alcántara Ávila, F., Rabault, J., Miró, A., Font, B., Lehmkuhl, O., and Vinuesa, R. (2025). Flow control of three-dimensional cylinders transitioning to turbulence via multi-agent reinforcement learning. *Communications Engineering*, 4(1). ([doi](#), [eprint](#))
4. Radhakrishnan, S., Calafell, J., Miró, A., Font, B., and Lehmkuhl, O. (2024). Data-driven wall modeling for les involving non-equilibrium boundary layer effects. *International Journal of Numerical Methods for Heat & Fluid Flow*, 34(8):3166–3202. ([doi](#), [eprint](#))
3. Varela, P., Suárez, P., Alcántara-Ávila, F., Miró, A., Rabault, J., Font, B., García-Cuevas, L.M., Lehmkuhl, O., and Vinuesa, R. (2022). Deep reinforcement learning for flow control exploits different physics for increasing reynolds number regimes. *Actuators*, 11(12). ([doi](#), [eprint](#))
2. Font, B., Weymouth, G.D., Nguyen, V.-T., and Tutty, O.R. (2021). Deep learning of the spanwise-averaged Navier–Stokes equations. *Journal of Computational Physics*, 434:110199. ([doi](#), [eprint](#))
1. Font, B., Weymouth, G.D., Nguyen, V.-T., and Tutty, O.R. (2019). Span effect on the turbulence nature of flow past a circular cylinder. *Journal of Fluid Mechanics*, 878:306–323. ([doi](#), [eprint](#))

Peer-reviewed symposium proceedings

4. Montalà, R., Font, B., Lehmkuhl, O., Vinuesa, R., and Rodriguez, I. (2026). High-lift wing separation control via Bayesian optimization and deep reinforcement learning. In *Journal of Physics: Conference Series*, volume 3230, page 012017. ([doi](#), [eprint](#))

3. Font, B., Alcántara-Ávila, F., Rabault, J., Vinuesa, R., and Lehmkuhl, O. (2024). Active flow control of a turbulent separation bubble through deep reinforcement learning. In *Journal of Physics: Conference Series*, volume 2753, page 012022. ([doi](#), [eprint](#))
2. Radhakrishnan, S., Gyamfi, L.A., Miró, A., Font, B., Calafell, J., and Lehmkuhl, O. (2021). A data-driven wall-shear stress model for les using gradient boosted decision trees. In *ISC High Performance Computing Conference*, pages 105–121. ([doi](#), [eprint](#))
1. Font, B., Weymouth, G.D., Nguyen, V.-T., and Tutty, O.R. (2020). Turbulent wake prediction using deep convolutional neural networks. In *33rd Symposium on Naval Hydrodynamics*. Office of Naval Research, US. ([eprint](#))

Conference proceedings

8. Font, B., Lauber, M., Huang, T.-Y., and Weymouth, G.D. (2026). Scaling WaterLily.jl with MPI and an improved geometric multigrid solver. In *ParCFD 2026, Nantes (France)*. ([eprint](#))
7. Montalà, R., Font, B., Suárez, P., Rabault, J., Lehmkuhl, O., Vinuesa, R., and Rodriguez, I. (2025). Discovering flow separation control strategies in 3D wings via deep reinforcement learning. In *15th ERCOFTAC Symposium on Engineering Turbulence Modelling and Measurements*. ([eprint](#))
6. Montalà, R., Font, B., Suárez, P., Rabault, J., Lehmkuhl, O., Vinuesa, R., and Rodriguez, I. (2025). Deep reinforcement learning for active flow control around a three-dimensional flow-separated wing at $Re = 1000$. In *1st International Symposium on AI and Fluid Mechanics*. ([eprint](#))
5. Cabral, M., Font, B., and Weymouth, G.D. (2024). ϕ -flow: A novel physics-constrained architecture to enforce incompressibility and boundary conditions for fast and accurate flow predictions. In *16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics*, WCCM 2024. CIMNE. ([doi](#), [eprint](#))
4. Montalà, R., Font, B., Lehmkuhl, O., Vinuesa, R., and Rodriguez, I. (2024). Towards active flow control strategies through deep reinforcement learning. In *16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics*, WCCM 2024. CIMNE. ([eprint](#), [url](#))
3. Suárez, P., Alcántara-Ávila, F., Miró, A., Rabault, J., Font, B., Lehmkuhl, O., and Vinuesa, R. (2023). Active flow control for three-dimensional cylinders through deep reinforcement learning. In *14th International ERCOFTAC Symposium on Engineering, Turbulence, Modelling and Measurements*. ([eprint](#))
2. Weymouth, G.D. and Font, B. (2023). WaterLily.jl: A differentiable fluid simulator in Julia with fast heterogeneous execution. In *ParCFD 2023, Cuenca (Ecuador)*. ([eprint](#))
1. Font, B., Weymouth, G.D., and Tutty, O.R. (2017). Analysis of two-dimensional and three-dimensional wakes of long circular cylinders. In *OCEANS 2017*. IEEE. ([doi](#), [eprint](#))

Invited Talks

20. Data-driven methods seminar at University of Manchester, UK (2026)
19. SciComp seminar series at RPTU University Kaiserslautern-Landau, Germany (2026). ([url](#))
18. Julia4PDEs workshop at the Vrije Universiteit, Netherlands (2026). ([url](#))
17. Dipartimento di Ingegneria e Scienze Applicate at University of Bergamo, Italy (2026). ([url](#))
16. Open-source and Benchmarking at the NUM-SCARS ERCOFTAC workshop at CWI, Netherlands (2026). ([url](#))
15. Computational Fluid Dynamics group visit at MARIN, Netherlands (2025)
14. AI and Deep Learning applied to Scientific Research at the Spanish Supercomputing Network Users Day, Spain (2025). ([url](#))
13. AI Meets Scientific Computing: Bridging Theory and Practice, University of Bergamo, Italy (2025). ([url](#))
12. Interdisciplinary Scientific Computing Laboratory, Pennsylvania State University, US (2024). ([url](#))
11. Fluids AI Special Interest Group series, TU Delft, Netherlands (2024)
10. Julia for HPC minisymposia, JuliaCon2024, Netherlands (2024). ([url](#))
9. Julia for HPC course, High Performance Computing Center Stuttgart (HLRS), Germany (2023)
8. PPPL Computer Science Department’s Machine Learning seminar, Princeton University, US (2021)

7. Engineering Mind Podcast (2021). ([url](#))
6. Applied Mathematics in Aerospace Engineering seminar, Universidad Politecnica de Madrid, Spain (2021)
5. Applied Math Colloquium, University North Carolina, US (2021)
4. Ocean Engineering, University Rhode Island, US (2021)
3. Fluid Dynamics Group at the Institute of High Performance Computing (A*STAR), Singapore (2020)
2. Fluid Structure Interactions Group Seminar series, University of Southampton, UK (2020)
1. Fluid Structure Interactions Group Seminar series, University of Southampton, UK (2017)

Grants

5. Physics and control of shock-wave/turbulent boundary-layer interactions (CO-PI). NWO Large Compute applications (2026). Awarded 72M CPU-hours on Snellius (Dutch national supercomputer)
4. Machine-learning based numerical methods for high-order scale-resolving simulations (PI). Incoming Visiting Researchers Program, University of Bergamo (2025). Awarded 2,800 euros to cover mobility expenses
3. FastFlows: Fast scale-resolving simulations of turbulent flows through physics-based machine learning. NWO Veni 2024 personal grant (single PI) (2025). Awarded 320,000 euros (NWO file number 21886). ([url](#))
2. High-lift wing separation control using deep reinforcement learning (co-pi). 6th Madrid Turbulence Summer Workshop (2025). Awarded 1-month funded project at the 6th Madrid Turbulence Summer Workshop. ([url](#))
1. DEAREL: Large-scale deep reinforcement learning for active flow control in wings (PI). EuroHPC Regular Access call (2024). Awarded 90,000 node-hours on MareNostrum5 (EHPC-REG-2024R01-038)

Reviewer

10. [Journal of Computational Physics](#)
9. [Journal of Fluid Mechanics](#)
8. [Nature Communications](#)
7. [Flow, Turbulence & Combustion](#)
6. [Journal of Wind Engineering & Industrial Aerodynamics](#)
5. [Nonlinear Dynamics](#)
4. [Ocean Engineering](#)
3. [Data-Centric Engineering](#)
2. [Results in Engineering](#)
1. [Engineering with Computers](#)

Student supervision

Mentored and supervised students at different stages of their educational program such as Undergraduate students, MSc students, and PhD students. As a mentor, the goal is to motivate students to pursue an interesting scientific topic while providing guidance throughout the process of learning and achieving.

PhD students

- C. Wang, Wall-modelled large-eddy simulation for wind-assisted ships, TU Delft 2025–
- P. Muñoz, Non-equilibrium wall modelling using machine learning, Barcelona Supercomputing Center 2024–
- M. Cabral, Physics-based machine learning of marine hydrodynamics, TU Delft 2023–

MSc projects

- Gaussian bump Bayesian optimization for turbulence transition control, TU Delft 2026
- Generative machine learning models for turbulent flow simulations, TU Delft 2026
- Prediction of hydrofoil ventilation onset using temporal forecasting, TU Delft 2026
- Ship motion forecasting by sea state observations, TU Delft 2026
- E. van der Weide, Numerical model of Sarsia tubulosa jellyfish, TU Delft ([url](#)) 2026
- E. Chillón, Acceleration of an AMG pressure solver using graph neural networks, TU Delft ([url](#)) 2025
- B.U. Ekizoğlu, Predicting wake-induced interactions of tandem Flettner rotors, TU Delft ([url](#)) 2025
- K. von Streit, Flow field prediction using physics based machine learning, University of Southampton 2020
- I. Elizalde Castells, Machine-learning wall model for bluff bodies forces calculation, University of Southampton 2019

Undergraduate projects

- I. Simó Muñoz, Discovering new scaling laws in turbulent boundary layers via multi-expression programming, Universitat Politècnica de Catalunya ([url](#)) 2021
- M. Tosquella Riera, Discovering new expressions for the vortex trajectories and velocity profiles of synthetic jets, Universitat Politècnica de Catalunya ([url](#)) 2021

Teaching

Lecturer at TU Delft. Also, served as demonstrator and marker of multiple modules during my PhD. Demonstration-related tasks involved preparing and delivering the laboratory sessions which included a theory part and an experimental part. Additionally, served as lecturer of the BSC summer school on AI and HPC delivering a lecture on AI for CFD.

- Lecturer at TU Delft 2024–
 - Instructor at Computational Fluid Dynamics for Engineers (ME45155): Machine learning for CFD
 - Instructor at Hydromechanics (MT2461): Hydrofoil lab, exam preparation, marking
 - Invited lecturer at Numerical Ship Hydrodynamics (MT44025): Turbulence modelling
- Lecturer at the PUMPS+AI Summer School, Barcelona Supercomputing Center 2022
 - Machine learning for computational fluid dynamics ([url](#))
- Demonstrator, University of Southampton 2015–2017
 - Aerodynamics: Nozzle lab
 - Propulsion: Ramjet, turbojet and rocket engine labs
 - Aerothermodynamics: Marking of lab reports

Software

Programming languages: Fortran (including MPI), Julia, Python (including PyTorch, Keras, and Tensorflow), C, Java, Matlab.

Tools: Git, L^AT_EX, Inkscape, Paraview.

Selection of popular repositories:

- Font, B. (2024). SmartSOD2D: A machine-learning framework for the SOD2D CFD solver using SmartSim. ([url](#))
- Weymouth, G.D. and Font, B. (2023). WaterLily.jl: Fast CFD Julia solver based in the finite-volume and immersed boundary methods with backend-agnostic execution (CPU and GPU). ([url](#))
- Font, B. (2020). SANSPy: Spanwise-averaged Navier–Stokes modelling through convolutional neural networks. ([url](#))

- Font, B. (2019). F2Py-Sockets: A minimal working example on how to use sockets to transfer data from Fortran to Python and viceversa. ([url](#))
- Font, B. (2018). NuatsBot: Technical analysis trading bot. ([url](#))
- Font, B. (2016). PostProc: A post-processing package for my PhD CFD simulations. ([url](#))

Open science statement

I advocate for open science. Most of my papers have an e-print version that can be downloaded for free either on [arXiv](#) or [my website](#). The software I develop is also open-source, and you can find it in my [Github repository](#).